



F107 – Foundations of Computer Science

Lecture 1

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Topics



- Foundations of Computer Science
 - Computation models
 - Computer components
 - Hardware vs. Software
 - History of computing machines
 - Types of computing machines
- Number systems
 - Decimal, binary, octal, hexadecimal systems
 - Basic arithmetic
- “Foundations of CS” by B. Forouzan
 - Chapters 1 and 2

Computer



- What is a computer?
 - Electronic device
 - Takes input (instructions from the user), process it and gives the desired output (result)

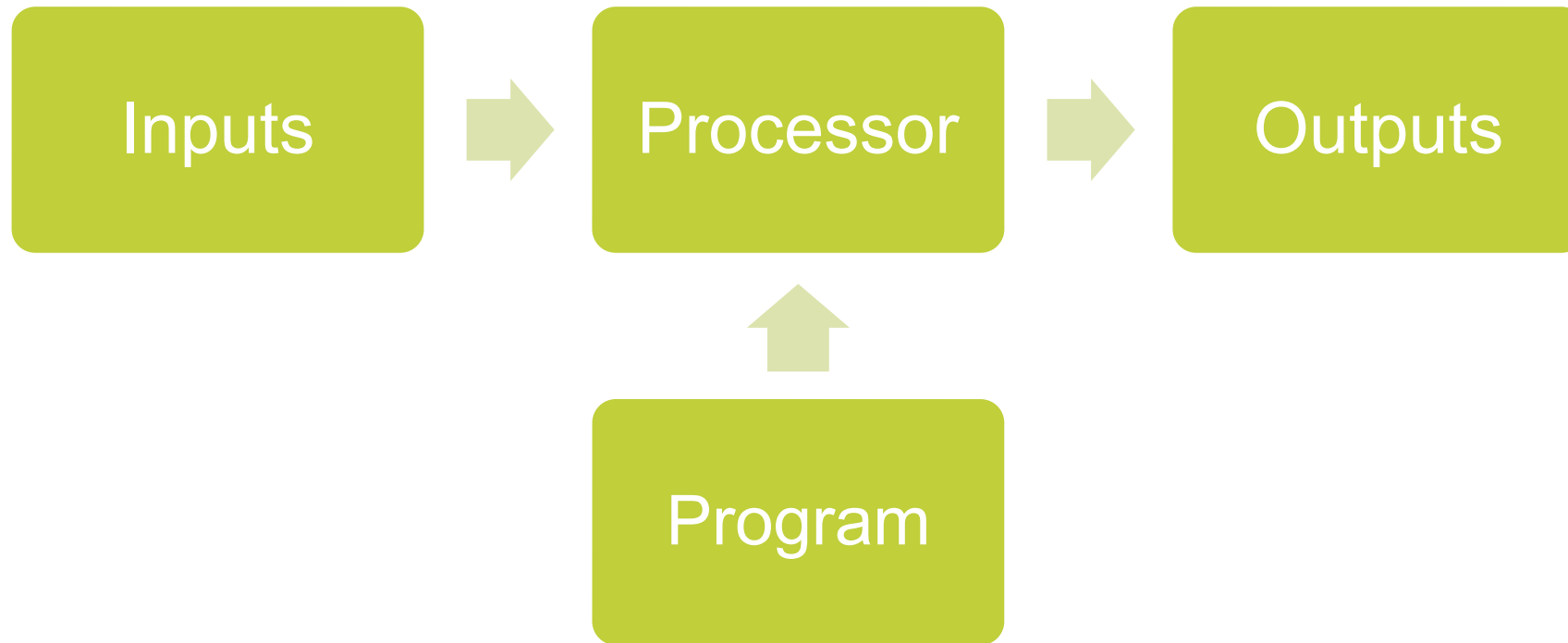
Computer

- What is a computer?

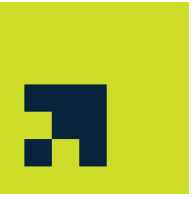


Computer

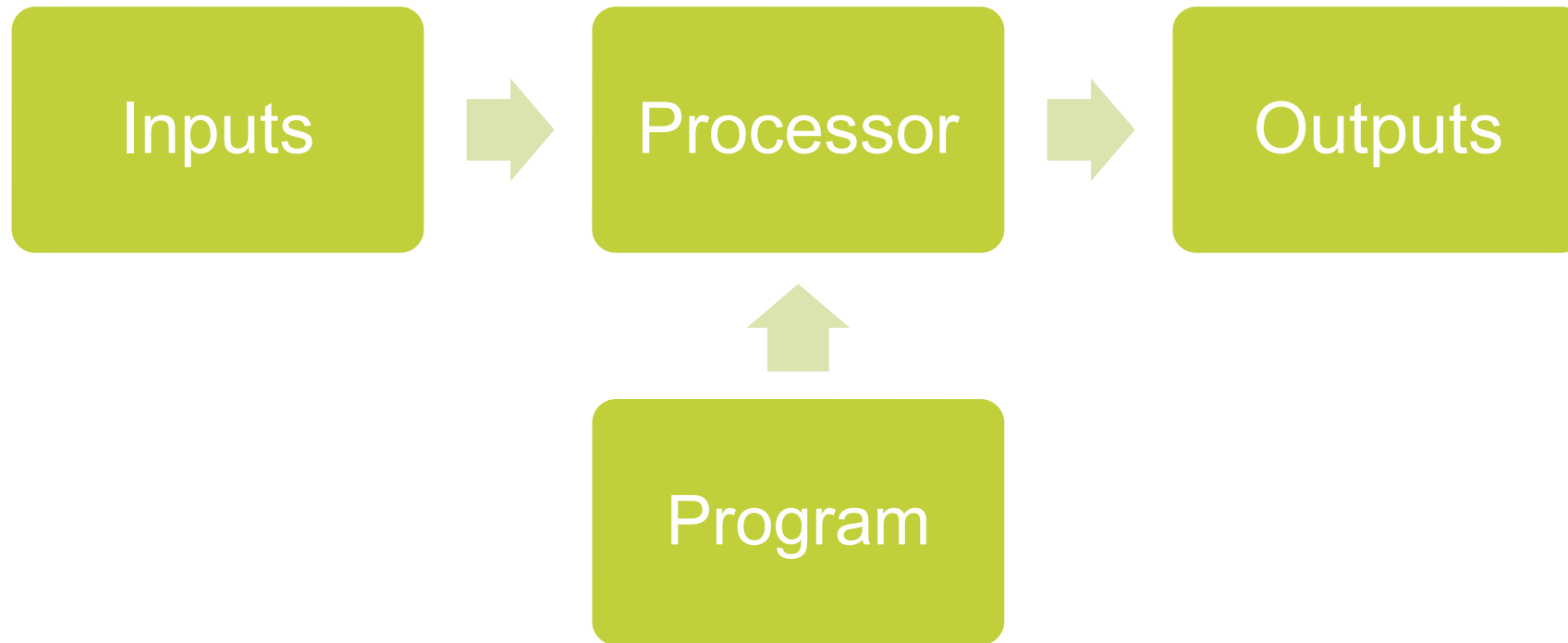
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Computer



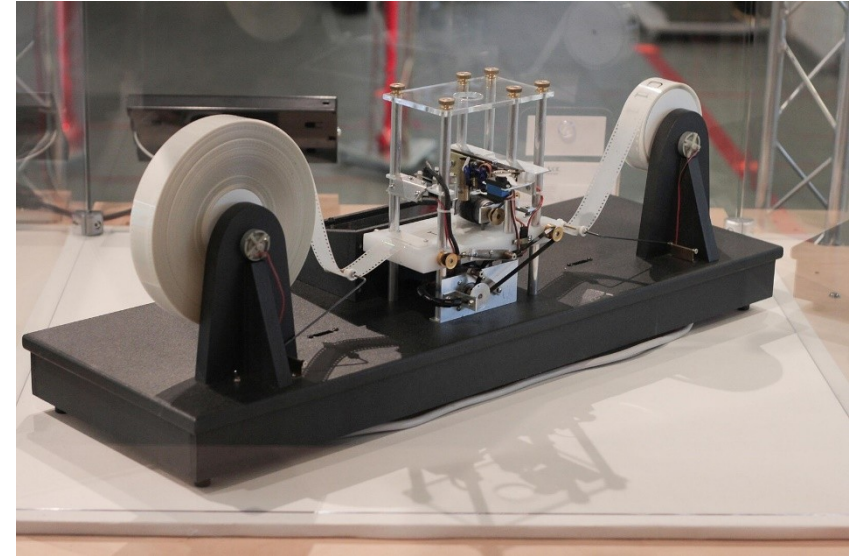
- What is a computer?



- Programmable processor that transforms inputs in well-defined manner into outputs

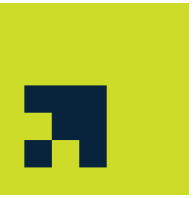
Turing Machine Model

- Formalization proposed by Alan Turing in 1936
 - Theoretical device that edits symbols on a tape
 - Defines computability
 - Basis for theoretical computer science and current computers
- Operation of a specific Turing Machine
 - Infinitely long memory tape
 - Discrete cells containing symbols from an alphabet
 - Machine has two major components
 - Head points at a cell
 - State
 - Actions:
 - Process content of cell, depending on current state
 - Go to left, go to right, or halt



				q_4			
	s_1	s_1	s_3	s_1	s_0	s_1	

Universal Turing Machine



- What about different tasks?
 - Tasks have an own associated Turing machine (or multiple)
- Generalization: Universal Turing Machine
 - A single machine that can simulate any other Turing machine
 - Interprets both as input
 - Description of a specific Turing Machine
 - Actual data to work on
 - Introduces programmability into theoretical computing
- Important ability: “Turing Completeness”
 - A “computing device” that can simulate any Turing Machine

Not all tasks can be
computed!

Von Neumann Architecture Model

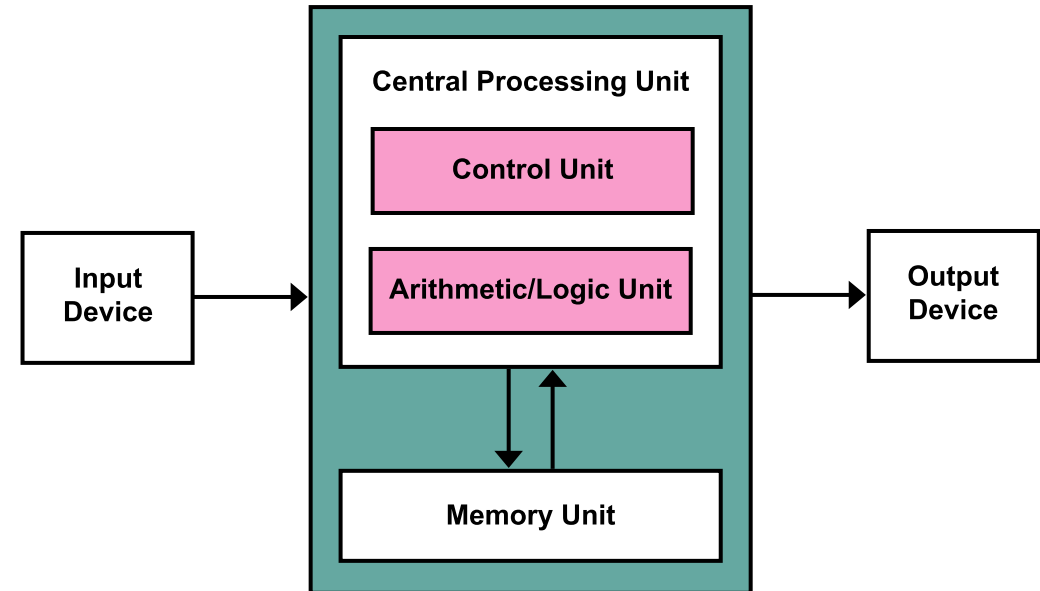


- Concept introduced in 1945 by John von Neumann
 - Foundation in most modern computer systems
 - Hardware vs. Software

Von Neumann Architecture Model



- Concept introduced in 1945 by John von Neumann
 - Foundation in most modern computer systems
 - Hardware vs. Software
- Computer consists of two major parts
 - Hardware
 - Memory
 - Central processing unit (control unit, ALU)
 - Input and outputs (I/O)
 - Software
 - Program and data
 - Both stored in the same memory
- Extensions of the V-N-A: Specialized hardware components

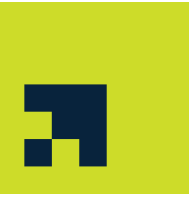


Components of a (Present-Day) Computer



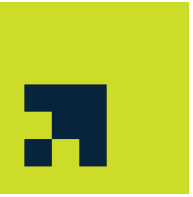
- Hardware: physical parts
 - Mainboard
 - CPU
 - Memory (RAM, HDD, SSD, ...)
- Software: operational instructions
 - Operating System
 - Programs
- Data:
 - Binary format representation of values

Programs and Algorithms



- Instruction
 - Most basic single task for a CPU
- Algorithms
 - Step-by-step methods to solve problems
- Programs
 - Sequences of instructions that realize algorithms
 - Instructions are grouped into functions
 - Allow to reuse sets of instructions to make programming scalable
 - Take inputs and generate outputs
- Programming language
 - Formal representation of instructions
 - Following specific syntax and have semantics
 - High-level and low-level languages

Taxonomy of Computers



- Taxonomies based on

Computational Capabilities

Supercomputers: extremely fast, parallel systems

Mainframes: handle large-scale processing

Minicomputers: devices support multiple concurrent users

Microcomputers: devices support one user; laptops, desktops

Embedded systems: specific-purpose devices

Purpose

General Purpose Computers:
can be tailored to needs

Special Purpose Computers:
Special setup for one task

Data Handling

Best-Effort System:
Unpredictable performance

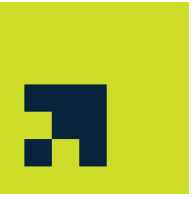
Real-Time System: Guaranteed
execution times

Location

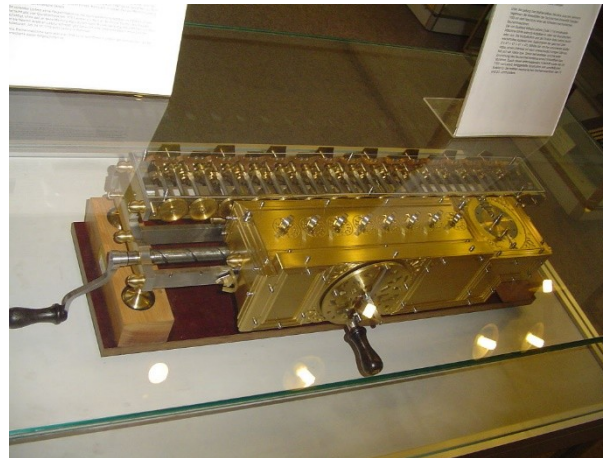
Local System: All components
locally available

Distributed System:
Geographically separation of
computation/storage

History of Mechanical Computing Machines



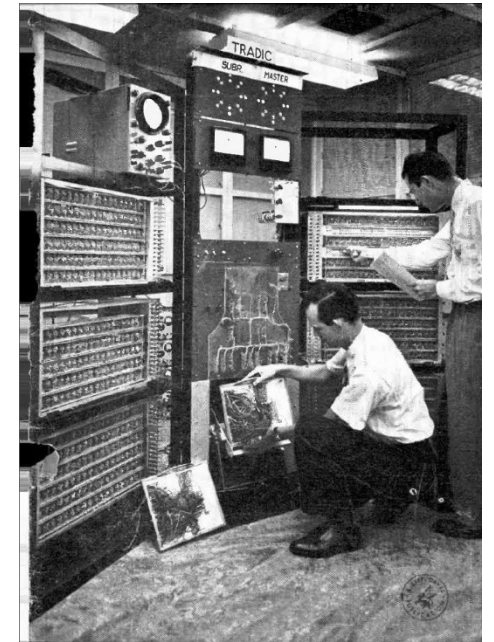
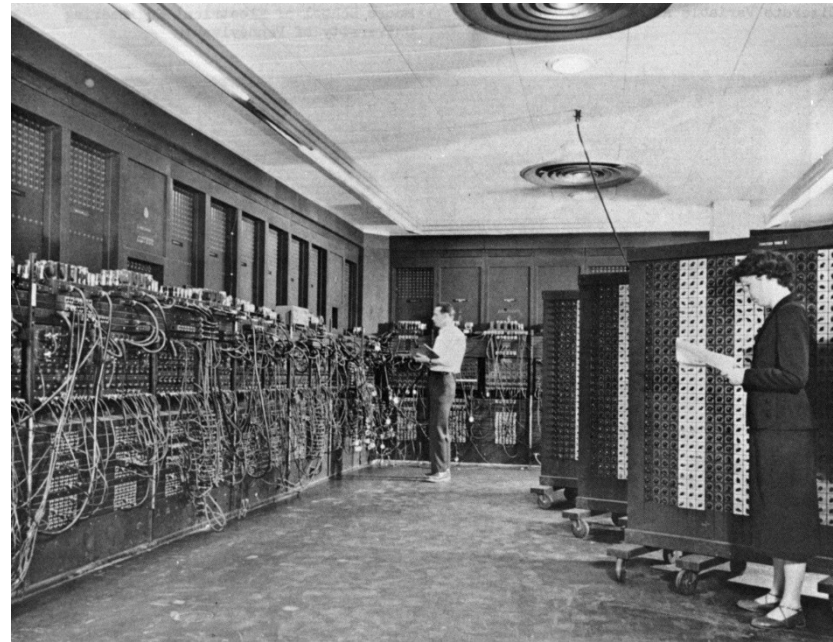
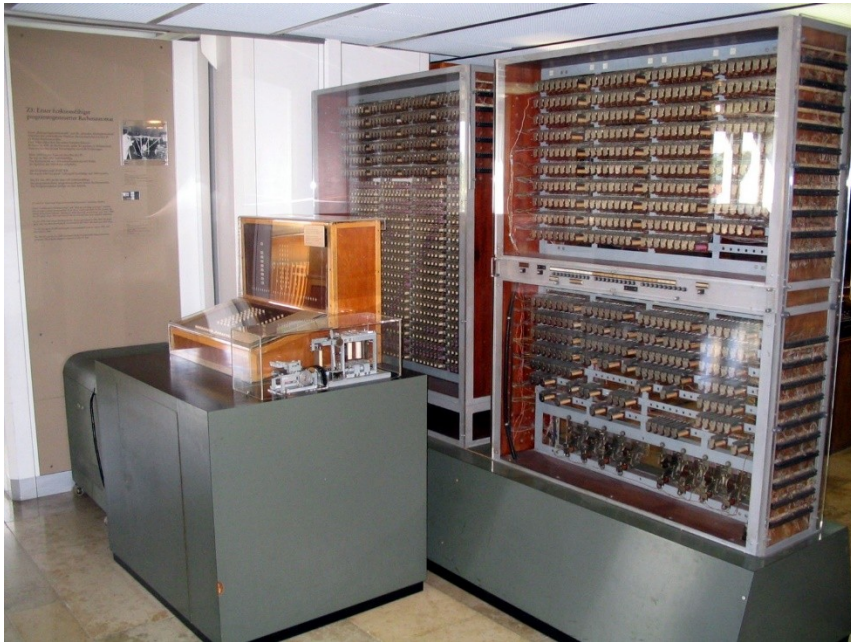
- Pascaline (1642) – mechanical adder (Pascal)
- Leibniz Wheel (17th century) – four-function calculator
- Jacquard Loom (1804) – punch card programming for looms
- Analytical Engine (1837) – Babbage created device that had ALU, memory, and control engine
- Enigma (1940) – Encryption and decryption machine for secure communication



History of Electronical Computing Machines



- Z3 (1941): Konrad Zuse invented first programmable electronic computer
- ENIAC (1945): first general-purpose computer
- EDSAC (1949): first to use stored program concept
- TRADIC (1954): first transistor-based computer



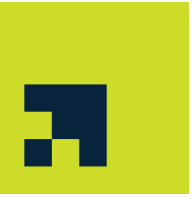
Number Systems Overview



- Number systems define representation of numbers using symbols
 - Positional systems
 - Non-positional systems
- Symbols represent certain meaning
 - Can be combined
- Humans: Decimal System
 - Base-10
- Machines: Binary System
 - Base-2

⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
0	1	2	3	4	5	6	7	8	9
○	᠑	᠒	᠓	᠔	᠕	᠖	᠗	᠘	᠙
·	᠑	᠒	᠓	᠔	᠕	᠖	᠗	᠘	᠙
○	一	二	三	四	五	六	七	八	九
零	壹	貳	叁	肆	伍	陆	柒	捌	玖

Decimal System (Base-10)



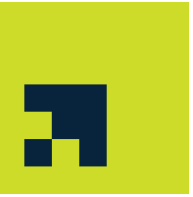
- Each position in a number is to Base-10
 - Base: 10
 - Alphabet: $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$
- Standard for human-centric numeric systems
- Example: $52_{10} = 5 \times 10^1 + 2 \times 10^0$
- Exercise: Rewrite the numbers
 - $9290_{10} =$
 - $56023_{10} =$
 - $2566221_{10} =$

Binary System (Base-2)



- Each position in a number is to 2
 - Base: 2
 - Alphabet: $\{0, 1\}$
- Standard for storage and processing in computer systems
- Example: $10110_2 = 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$
- Exercise: Rewrite the numbers
 - $0_2 =$
 - $111111_2 =$
 - $101011110000_2 =$

Octal and Hexadecimal System (Base-8/-16)



- Octal system: Each position in a number is to Base-8
 - Base: 8
 - Alphabet: {0, 1, 2, 3, 4, 5, 6, 7}
- Hexadecimal system: Each position in a number is to Base-16
 - Base: 16
 - Alphabet: {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F}
- Compact representations of binary data
- Examples:
 - Octal: $7412_8 =$
 - Hexadecimal: $42BC2FA_{16} =$

Conversions to Base-10



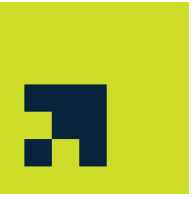
- Use the previous representation and add up the values
- Exercise: Transform the following numbers to Base-10 representation
 - $1101_2 =$
 - $043_8 =$
 - $3210_8 =$
 - $1234_{16} =$
 - $100F_{16} =$

Conversions from Base-10



- Division with remainder (rest)
 1. Divide number given in Base-10 by new base
 2. Write down remainder of the division, even if 0 (from right to left)
 3. If the whole part of the division is not zero, go back to Step 1; if it is zero, stop
- Example: $561_{10} = ?_2 = ?_{16}$

Real Numbers vs. Integer Numbers



- Integers $\mathbb{Z}$: No fractional part
- Reals $\mathbb{R}$: Include fractional part
- How to represent fractions? → Different format required
 - Precision
 - Storage
 - Computational effort
- Reals are represented as “floats” (IEEE 754)
 - Floating-Point and Fixed-Point Notations
 - Example: 1001.011_2

See Session 3

Arithmetic in Base-2



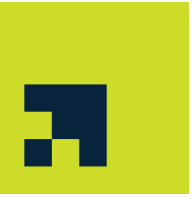
- Arithmetic works in any base the same
 - Overflows occur, but must respect the new base
- Addition: $90_{10} + 54_{10} = ?_{10}$

Arithmetic in Base-2



- Multiplication: $56_{10} \times 84_{10} = ?_{10}$

Non-Positional Number Systems



- Symbols have fixed values regardless of position
 - Some inherent conventions to represent numbers
 - Less compact and less scalable
 - Used historically before positional systems

- Example: Roman numerals

- I = 1, III = 3, IV = 4, XLIV = 44
- MMXXV =

I	V	X	L	C	D	M
1	5	10	50	100	500	1000

- Extra: Example: Mayan system



Task 1



- Explain the key differences between the Turing and Von Neumann models

Task 2



- Create a taxonomy diagram showing types of computers with examples

Task 3



- Describe the historical significance of the Jacquard Loom

Task 4



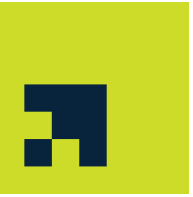
- List and describe the four components of the Von Neumann architecture

Task 5



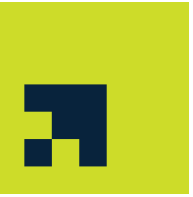
- What makes a system 'Turing complete'?

Task 6



- Transform the following numbers to Base-10 representation
 - $1101110_2 =$
 - $1010110011110000_2 =$
 - $01243_8 =$
 - $7772222_8 =$
 - $97141AFFE1234_{16} =$
 - $DEADCAFE_{16} =$

Task 7



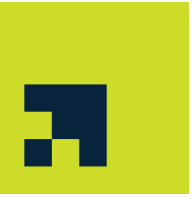
- Transform the following numbers to Base-2, Base-8, and Base-16:

- $590_{10} =$

- $9999_{10} =$

- $1000000_{10} =$

Task 8



- Add and multiply the following two numbers in Base-10 and in Base-2

- $567_{10} + 845_{10} =$

- $567_{10} \times 845_{10} =$

Task 9



- Describe the steps to convert a fractional Base-2 number to Base-10

Task 10



- Describe floating-point representation and mention its limitations